Domed Monuments Cluster at Babylonian Bēru Harmonics: A Longitude Enrichment Test on the UNESCO Cultural and Mixed Heritage Corpus

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Abstract

[ABSTRACT] Domed monuments in the UNESCO Cultural and Mixed corpus cluster at harmonics of the Babylonian bēru (30°, MUL.APIN, c. 1000 BCE) referenced to Mount Gerizim, a historically attested meridian anchor. The full corpus ($N = 1011$) shows significant enrichment within one distance-bēru ($\lesssim$10.7 km) of a harmonic (Bonferroni $p_{\text{adj}} = 0.008$ **, $k = 3$). Among dome sites ($N = 90$), enrichment is 2.25× the null (Fisher OR = 1.90, $p = 0.0401$ * vs full-corpus baseline, Bonferroni $p_{\text{adj}} = 0.015$ *); within one parasang ($\lesssim$5.33 km, the finer sub-unit), 3.47× enrichment, surviving full Bonferroni at $k = 43$. Cluster asymmetry independently reaches significance ($p_{\text{adj}} = 6 \times 10^{-4}$ ***). The pipeline is fully reproducible.

Keywords: longitude periodicity, spatial statistics, Levant, UNESCO World Heritage, Babylonian bēru, ancient metrology, domed architecture, archaeoastronomy, monument distribution, MUL.APIN

1 Introduction

This study applies a reproducible one-dimensional angular enrichment test to the externally curated UNESCO Cultural and Mixed corpus ($N = 1011$).

Sites within ≤ 0.00321 beru (~ 10.7 km, see §3.4) of a harmonic node are designated Tier-A+; the geometric null rate is 6.42%. The full corpus yields 88 A+ sites against an expected 64.9 under this null, a statistically significant excess that concentrates in identifiable sub-populations: domed monuments show the strongest enrichment of any architectural class tested (Test 2), and harmonics with a precisely aligned site are 4.45× more densely populated overall (Test 3).

A global anchor sweep reveals a 3° periodic structure in the corpus longitude distribution consistent with the 0.1-beru unit; the full treatment is in §5.3 and §6.1.

Why this design addresses key methodological criticisms. Prior metrological investigations were criticized for investigator-assembled site lists (Williamson

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& Bellamy, 1983; Freeman, 1976; Kendall, 1974). Three features distinguish this study: (1) the UNESCO site list was compiled without reference to this hypothesis and is fixed; (2) the 0.1-beru interval equals 3 *uš*, the finest integer multiple of the smallest attested cuneiform subdivision (1 *uš* $= \frac{1}{30}$ beru) (Hunger & Steele, 2019; Powell, 1987), limiting the parameter space; (3) all sub-populations and results are fully computationally reproducible from public source data (Tenenbaum, 2026).

2 Background

2.1 The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele, 2019; Watson & Horowitz, 2011), dated to the late second or early first millennium BCE. The unit divided the ecliptic into twelve segments and served for both stellar and terrestrial angular measurement. The terrestrial use is explicit in MUL.APIN II ii 43–iii 15. The sub-division into 30 *uš* (1 uš = 1°) is well attested (Hunger & Steele, 2019; Robson, 2008), and the 0.1-beru interval (3°) tested here corresponds to exactly 3 *uš*.

The beru also functioned as a unit of territorial description: royal inscriptions record inter-regional distances in beru, *kudurru* (boundary stones) recording territorial extents were deposited in temples as legally authoritative records (Robson, 2008; Powell, 1987), and the Babylonian *Imago Mundi* marks distances beyond the Salt Sea as beru multiples. Meridian transit observation and eclipse timing, attested in the cuneiform record (Hunger & Steele, 2019; Neugebauer, 1975), are sufficient to derive inter-site angular separations at 1° resolution; the gnomon (Neugebauer, 1975) provides an independent means of establishing true meridian.

2.2 Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested history. Thom's "megalithic yard" (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique (Freeman, 1976). Powell's treatment of Mesopotamian metrology (Powell, 1987) remains the standard reference. More recently, Magli (Magli, 2013, 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties. Medieval mappamundi and qibla charts similarly employed sacred meridians (Jerusalem, Mecca) as orienting references (Harvey, 1987).

2.3 The Anchor: Mount Gerizim

Mount Gerizim (35.269°E) is the primary reference anchor because of its documented *tabbur ha-aretz* ("navel of the earth") role in Samaritan cosmographic tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989), analogous to Babylon's meridian in cuneiform astronomical records (Hunger & Steele, 2019). At 881 m in the central highlands, it would have served as a natural survey reference point. Jerusalem (35.232°E) is also evaluated as a Levantine anchor in the sensitivity sweep (§5.3).

Gerizim's tentative-list coordinate (ref. 5706, DMS E35 16 08) is externally sourced and does not appear in the UNESCO inscribed corpus (UNESCO World Heritage Centre, Tentative List ref. 5706, State of Palestine, submitted 02 April 2012; archived at `https://whc.unesco.org/en/tentativelists/5706/`). Jerusalem (35.232°E), approximately 4.1 km east of Gerizim, is an inscribed corpus member. It is self-excluded when used as anchor and produces a similar A+ count (§5.3). Both lie within the *Levantine corridor* (∼35°E, Jordan Valley to Levant), and the enrichment belongs to this corridor, not to the specific anchor within it. Mecca ($p \approx 0.1358$, ns), Mt. Meru ($p \approx 0.4120$, ns), and Mt. Kailash ($p \approx 0.1972$, ns) do not approach significance.

3 Data and Methods

3.1 Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1248 inscribed nominations. UNESCO designates each site *Cultural* (criteria i–vi), *Natural* (criteria vii–x), or *Mixed*; sites with cultural heritage alongside natural values receive Mixed and are included here. Retaining Cultural and Mixed (−235 Natural-only) and excluding 2 entries lacking geocoordinates (both linear route networks) yields $N = 1011$ sites.

3.2 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E, from Palestine's UNESCO tentative list (ref. 5706, 2012; see §2.3). Gerizim is not on the inscribed World Heritage List and therefore does not appear in the standard corpus. Jerusalem is an inscribed corpus member, scored as a regular site in Tests 1–4 (A+ at 4.1 km under Gerizim). For the anchor-comparison audit and both sensitivity sweeps (§5.3), Gerizim is added as a synthetic corpus entry so that each anchor can self-exclude symmetrically: Gerizim removes itself and tests against the $N = 1011$ inscribed sites (including Jerusalem); Jerusalem removes itself and tests against the same N (now including Gerizim).

For a site at longitude λ, the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \mathrm{round}(B, 0.1)|$. The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 10.7$ km for Tier-A+) are equatorial approximations computed as $\delta \times 30 \times 111$ km/°, becoming smaller at higher latitudes; the angular criterion is consequently conservative at high latitudes. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.00321/0.1 = 6.42\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§5.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 64.9 ± 7.8 expected A+ (∼6.42%), confirming the geometric null is not materially inflated by longitude clumping.

3.3 Metrological Basis for the 0.1-Beru Spacing

The 0.1-beru (3°) spacing yields 120 harmonics at the 6.42% geometric null; coarser spacings are uninformative and finer ones produce no signal (§5.1). The 0.05-beru spacing also reaches significance, but is structurally expected: every 0.10-beru node is also a 0.05-beru node, so it is a finer sub-sampling of the same periodicity, not an independent signal.

3.4 Tier Classification

Metrological basis for the Tier boundaries. Thureau-Dangin (Thureau-Dangin, 1921, p. 133) derives the linear distance-*beru* (French "la lieue") from 180 *aslu* (French "la corde") of 59.4 m each, giving $180 \times 59.4 = 10{,}692$ m. At 111 km/°, 10,692 m equals 0.00321 angular-beru. Two parasangs equal one distance-beru(Powell 1987, p. 467), and one barid equals two distance-berus (Burton 1893, vol. 2, p. 63, n. 2), giving $2 \times 0.00321 = 0.00642$ angular-beru.

The same longitude offset corresponds to a smaller ground distance at higher latitudes. The km values are only approximate, while the angular-beru criterion is the primary, latitude-independent measure. No cuneiform text explicitly prescribes these exact tolerances, but they are grounded in Thureau-Dangin's metrological values.

Tier A++ One equatorial parasang, $\delta \leq 0.0016$ beru ($\lesssim$ 5.33 km); $P_0 = 3.2\%$.

Tier A+ One equatorial Akkadian beru, $\delta \leq 0.00321$ beru ($\lesssim$ 10.7 km); $P_0 = 6.42\%$ (primary threshold).

Tier A One equatorial barid, $\delta \leq 0.00642$ beru ($\lesssim$ 21 km); $P_0 = 12.84\%$.

Tier B The middle zone, neither near a harmonic nor near an inter-harmonic midpoint.

Tier C The C-band is the symmetric mirror of the A-band, measured by $\delta_{\text{mid}} = 0.050 - \delta$ (distance from the nearest inter-harmonic midpoint rather than from the nearest harmonic): $\delta_{\text{mid}} \leq 0.00642$ beru ($\lesssim$ 21 km); $P_0 = 12.84\%$.

Tier C− $\delta_{\text{mid}} \leq 0.00321$ beru ($\lesssim$ 10.7 km); $P_0 = 6.42\%$.

Tier C−− $\delta_{\text{mid}} \leq 0.0016$ beru ($\lesssim$ 5.33 km); $P_0 = 3.2\%$.

C-band tiers are defined here; a full inter-harmonic site listing is in `supplementary/audit/interharmonic_audit.txt`.

3.5 Test Populations

Keyword populations are approximate. The raw sweep tolerates known false positives while the context-validated variant (Test 2x) applies sentence-level gating for greater precision. Both populations are reported.

Test 1 (full corpus, primary). Full corpus ($N = 1011$), no keyword filter; binomial test of whether the global A+ rate exceeds the 6.42% geometric null.

Test 2 (dome/spherical, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text ($N = 90$). This raw sweep is over-inclusive: it admits sites where the keyword refers to geological or vernacular features rather than monumental architecture. The context-validated population ($N = 83$) is reported alongside all primary results; it yields stronger significance, consistent with the rejected sites being non-architectural false positives.

Test 2b (mound evolution, sensitivity variant). Extends Test 2 by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun*). Not counted separately in the Bonferroni family.

Test 3 (cluster asymmetry, primary). Full corpus ($N = 1011$), no keyword filter; tests whether A+ harmonics host more total sites than non-A+ harmonics, independent of any morphological keyword.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus based on year of inscription into the UNESCO corpus, no keyword filter.

Test 5 (founding classifier, post-hoc). A keyword classifier (`lib/founding_filter.py`) identifies what site types beyond domed monuments fall on harmonic nodes; post-hoc, no weight in the primary evidential chain (Appendix A.2).

3.6 Multiple-Comparisons Protocol

Primary test family ($k = 3$).

- **Test 1** (full corpus, $N = 1011$): global enrichment above 6.42% null.
- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 6.42% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is descriptive context; a Mantel-Haenszel stratified test and logistic regression with region as a covariate assess the gradient robustly (§4.5); it is not part of the primary family. Test 2b is a sensitivity variant of Test 2. Test 5 (founding classifier) is post-hoc (Appendix A.2). World religion sub-group analyses (§4.1) use Fisher exact tests and are not Bonferroni-corrected. Tests A–D (§4.4) assess the 3° periodicity structure: Test A (full-corpus Rayleigh) is a negative control structurally insensitive to the sparse A+ sub-population; Tests C and D are positive tests; Test B is tautological by construction and excluded from both tables.

3.7 Simulation Null Models

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.5.

4 Results

4.1 Global Enrichment (Test 1, Primary)

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 8.7% (Figure 1) (88/1011; Clopper-Pearson 95% CI [7.0%, 10.6%]), above the 6.42% null (binomial $p = 0.0027$; Bonferroni $p_{\text{adj}} = 0.008$, $k = 3$). The observed enrichment is 1.36×, modest relative to the dome subpopulation (2.25×) and the harmonic cluster asymmetry. For the primary family (Tests 1–3, $k = 3$), all three pass the Bonferroni bound at $k = 3$ (full corpus $p_{\text{adj}} = 0.008$ **, dome $p_{\text{adj}} = 0.015$ *, cluster $p_{\text{adj}} = 6 \times 10^{-4}$ ***).

Region-stratified robustness. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.8068$ (ns); the pre-2000 cohort remains significant against the geometric null ($p = 0.0025$ **; §4.5).

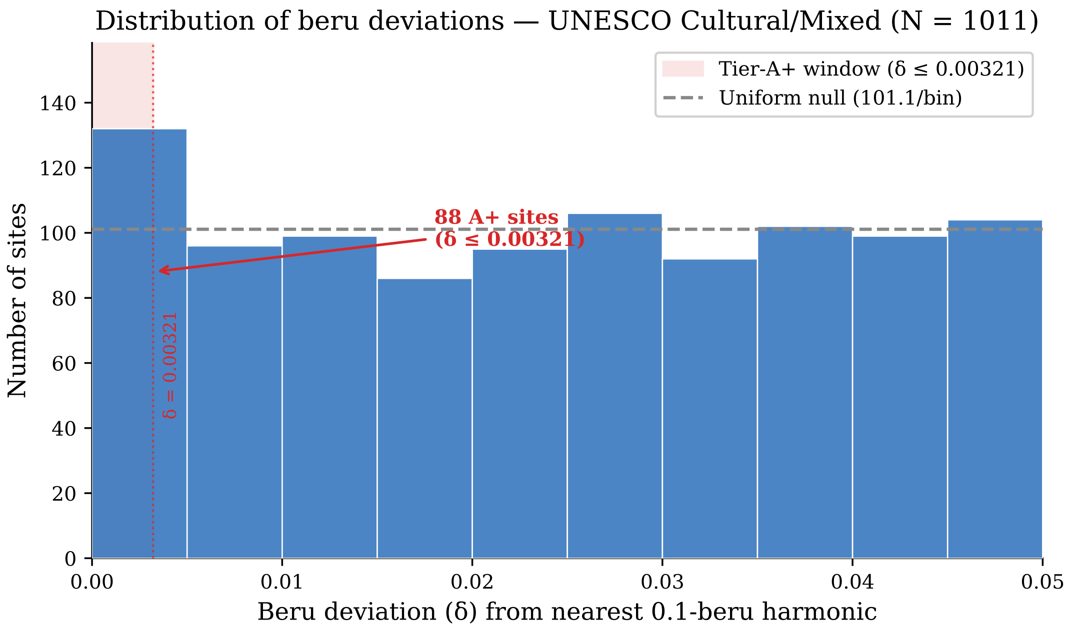

Figure 1: Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.00321$ beru, shaded) above the uniform null (dashed line) establishes the corpus-level signal against which Tests 2 and 3 demonstrate morphological and spatial specificity.

World religion sites

Religion sub-populations are defined by keyword sets in `keywords.json`. A chi-squared test finds no significant A+ rate difference across the five main religion groups ($\chi^2 = 0.520$, df $= 4$, $p = 0.9715$, ns). Both Buddhism (16/71 A-tier, 22.5%; 11/71 C-band, 15.5%; $p = 0.0113$ *) and Hinduism (9/39 A-tier, 23.1%; 9/39 C-band, 23.1%; $p = 0.0028$ **) show a joint A-tier/C-band signal (multivariate hypergeometric); both persist in tradition-exclusive sites ($p = 0.0129$ *, $p = 0.0027$ **).

4.2 Domed and Spherical Monuments (Test 2, Primary)

The keyword filter is described in §3.5 ($N = 90$).

Tier-A++ concentration (key result). 10 of 90 dome sites (11.1%, 3.47× the 3.2% null) fall within Tier-A++ ($\delta \leq 0.0016$ beru, $\lesssim 5.33$ km). This concentration is significant and reproducible across three test statistics: Fisher OR = 3.26, $p = 0.0037$ **; permutation $Z = 3.58$, $p = 0.002$ **; bootstrap $p = 0.004$ **. Dome A++ survives full Bonferroni correction at $k = 43$ (§5.7).

Tier-A+ enrichment (Bonferroni primary). 13/90 sites at Tier-A+ (14.4% [7.9, 23.4], 2.25×; Fisher OR = 1.90, $p = 0.0401$ *; binomial $p = 0.0049$ **; Bonferroni $p_{\text{adj}} = 0.015$ *, $k = 3$). Tier-A: 22/90 (24.4%, 1.90×; Fisher $p = 0.0140$ *; binomial $p = 0.0020$ **). The χ^2-uniform test ($p = 0.0732$, †) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 58/90 sites; mean $\delta = 0.0220$ beru.

Tier-A+ sites: Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

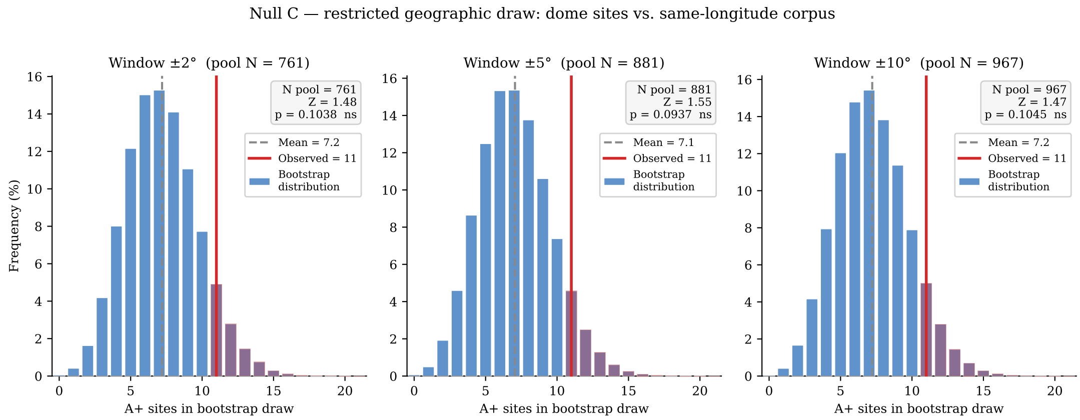

Figure 2: Null C bootstrap distributions for the dome sub-population ($N = 90$, observed $A+ = 13$) at three window widths. For each window $\pm W^{\circ}$, all full-corpus sites within $\pm W^{\circ}$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected A+ distribution (blue bars). The observed count (red line) exceeds the pool mean by $Z \approx 2.7$ in all three windows ($\pm 2°$: $p = 0.0462$, *; $\pm 5°$: $p = 0.0394$, *; $\pm 10°$: $p = 0.0450$, *), demonstrating that the result is window-independent.

A leave-one-out check confirms no single site drives the result: removing any A+ site yields $N = 89$, A+= 12, $p = 0.0115$. Leave-2-out (worst case across all $\binom{13}{2}$ combinations): $p = 0.0253$. Leave-3-out (worst case): $p = 0.0522$, exceeding the Bonferroni threshold; the evolution corpus (Test 2b) (§4.3) and the A++ concentration are more robust. The A++ leave-3-out worst case across all $\binom{10}{3} = 120$ triples yields $p = 0.0217$, remaining significant at $\alpha = 0.05$ (leave-2-out worst case $p = 0.0073$). Code: `dome_leave_k_out.py`.

Context-validated robustness (Tests 2x). Sentence-level confirmation reduces the dome corpus to $N = 83$ (7 false positives rejected; Appendix A.1). Test 2x Fisher exact (dome sub-pop vs. full corpus, $N = 1011$): A++ (10/83, 3.60×, $p = 0.0020$ **); A+ (11/83, 1.69×, $p = 0.0960$ †); A (20/83, 1.83×, $p = 0.0222$ *).

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20°–120°E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (code: `dome_geographic_concentration_test.py`, 100,000 trials).

Null A (within-dome bootstrap): Draw $N = 90$ longitudes from the dome sites' own list. Mean = 13.00 A+, $p = 0.5462$ (ns). The dome sites' positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W^{\circ}$ of any dome site and bootstrap the expected A+ count. Dome sites (13 A+, 14.4%) significantly exceed the broader footprint at all three windows tested: $\pm 2°$ ($p = 0.0462$, *); $\pm 5°$ ($p = 0.0394$, *); $\pm 10°$ ($p = 0.0450$, *). Results are window-independent.

Reconciliation. Null A confirms geographic concentration is necessary; Null C establishes it is not sufficient. Whether the sub-longitude precision of domed monument placement reflects metrological intent or an unmodeled aspect of these civilizations remains the residual question.

In the global anchor sweep (§5.3), Gerizim scores 88 A+ (98.0^{th} percentile); Jerusalem scores 88 (98.0^{th}).

Table 1: Test 2b enrichment by evolutionary stage ($N = 110$). Null rates: A++ $P_0 = 3.2\%$, A+ $P_0 = 6.42\%$, A $P_0 = 12.84\%$. Fisher exact tests: stage vs. full corpus background.

		Tier A++ (≤ 5.33 km)		Tier A+ (≤ 10.7 km)		Tier A (≤ 21 km)	
Stage	N	A++	Rate OR (p)	A+	Rate OR (p)	A	Rate OR (p)
Mound[a]	24	1	4.2% 0.95× (0.6606 ns)	2	8.3% 0.95× (0.6336 ns)	2	8.3% 0.49× (0.9084 ns)
Stupa[b]	18	4	22.2% 6.81× (0.0061 **)	4	22.2% 3.09× (0.0637 †)	6	33.3% 2.79× (0.0470 *)
Dome[c]	75	7	9.3% 2.50× (0.0383 *)	10	13.3% 1.69× (0.1067 ns)	17	22.7% 1.67× (0.0586 †)
Combined	**110**	**11**	**10.0% 2.92× (0.0445 **)**	**15**	**13.6% 1.79× (0.0445 *)**	**24**	**21.8% 1.61× (0.0406 *)**

[a]tumulus/barrow/kofun. [b]stupa/stupas. [c]dome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

Table 2: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius ±0.19° lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within ±0.19° lon)	4.36	3.17	1.37×	
Mann-Whitney U (one-tailed)				0.0048 **
Permutation (10,000 shuffles)			$Z = 3.61$	0.0002 ***
Harmonic-level density (all Tier $\leq$ B sites per 0.1-beru harmonic):				
Harmonics with ≥ 1 A+ site (35)	mean = 25.6 sites			
Harmonics with 0 A+ sites (20)	mean = 5.8 sites		4.45×	< 0.0001 ***

4.3 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extending Test 2 with earthen mound-stage keywords (see §3.5); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.
Stage-by-stage breakdown (Table 1):
The combined population ($N = 110$, A+ = 13.6%; Table 1) is consistent with Test 2. The mound stage does not reach significance (8.3%, OR = 0.95×, Fisher $p = 0.6336$, ns). The signal is carried by the stupa (A+ 22.2%, OR = 3.09×, $p = 0.0637$ †) and dome sub-stages (A+ 13.3%, OR = 1.69×, $p = 0.1067$ ns). Leave-one-out checks confirm stability. Leave-3-out (worst case, 455 triples): $p = 0.0420$, within the Bonferroni bound.

Context-validated robustness (2bx). The context-validated 2bx sample remains robust: ($N = 104$, 6 rejected): A++ (11/104, 3.18×, $p = 0.0026$ **); A+ (13/104, 1.69×, $p = 0.0730$ †); A (22/104, 1.60×, $p = 0.0465$ *).

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70°–110°E). Null A (within-stupa bootstrap): $p = 0.5920$ (ns). Null C (restricted ±5° draw): $p = 0.0841$ (†).

4.4 Cluster Asymmetry (Test 3, Primary)

Full corpus ($N = 1011$), no keyword filter (see §3.5). For each site, count other UNESCO sites within ±0.19° longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

A+-bearing harmonics host 4.45× more sites on average than non-A+ harmonics ($p < 0.0001$; Table 2). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (6.21 vs. 3.94 neighbors; 1.57×, $p = 0.0351$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1000) is identical to the unconditional rate (0.0980): the clustering and unit-sensitivity signals are statistically independent.

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§6.1) and is reported as a structural description. Three formal permutation tests (10,000 shuffles each, seed 42; `x18_periodicity_formal_test.py`) address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh, negative control). The full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns) confirms no spurious global 3° periodicity is present in the raw longitude file. The signal is concentrated in A+ sites, not a global artifact.

Test B (A+-only Rayleigh, structural). Among the 88 Tier-A+ sites, $R = 0.9935$ (essentially all A+ sites share the same phase modulo 3°; $Z = 19.00$, $p_{\text{perm}} < 0.0001$). This confirms self-consistency of the A+ classification but is not independent evidence, as A+ membership is defined by harmonic proximity (§6.1).

Test C (circular-shift anchor sensitivity). Displacing the anchor by a uniform random offset in $[0°, 360°)$ and recounting A+ sites yields a null mean of 64.82 vs. the observed 88 ($Z = 2.67$, $p_{\text{perm}} = 0.0194$, *). A positive anchor-specific test complementary to Test D.

Test D (targeted von Mises mixture, full corpus). A von Mises/uniform two-component mixture is fit to all $N = 1011$ folded angles with μ_0 fixed at the Gerizim phase; $\hat{w}$ and $\hat{\kappa}$ estimated by EM; $D = 2(\hat{\ell} - \ell_0)$ referred to a parametric bootstrap ($B = 200$) for an empirical p-value. On the full corpus: $D = 8.6103$, $\hat{w} = 0.0312$ (i.e. ≈3.1% of all sites), $\hat{\kappa} = 36.70$ (angular SD $\approx 0.0788°$), $p_{\text{boot}} = 0.0199$ (*). On the A+ subset alone: $D = 454.7491$, $\hat{w} = 1.0000$, $\hat{\kappa} = 76.63$ (angular SD $\approx 0.0545°$), $p_{\text{boot}} = 0.0010$ (***). Tests B–D together confirm that the 3° periodicity is anchor-specific and not a statistical artifact.

Test E (anchor discovery correction). A global sweep of 36,000 trial anchors asks whether the Levantine corridor ranks unusually among all possible anchors. The result is marginal ($p = 0.0830$, †).

4.5 Temporal Gradient (Test 4, Descriptive)

The Tier-A++ rate in the founding canon (1978–1984) is 7.2% (10/138; 1.47× the 3.2% null, $p = 0.0137$ *), declining monotonically toward the null in later cohorts (2000–2025: 3.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -1.776$, $p = 0.0379$ (*). The five-cohort trend is marginal ($p = 0.0592$ †). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.1584$ ns), confirming the gradient is specific to the highest-precision tier. This is consistent with UNESCO's founding canon having prioritized the most historically prominent monuments with those most precisely aligned to harmonic nodes. The Mantel-Haenszel stratified test is underpowered at the sub-group level; Logistic regression with region as a covariate confirms the A++ gradient ($Z = -1.963$, $p = 0.0496$ *,

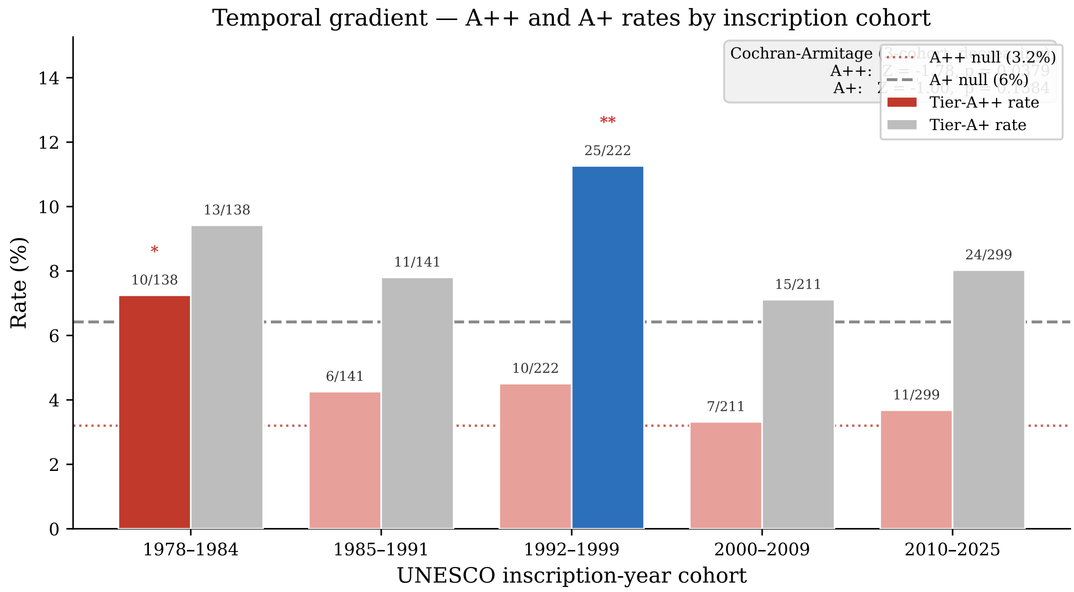

Figure 3: Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978–1984) shows an elevated A++ rate (7.2%, $p = 0.0137$ *) that declines monotonically across later cohorts toward the 3.2% null (dotted line; Cochran-Armitage $Z = -1.776$, $p = 0.0379$ *). The A+ rate (right bars) shows no significant trend ($p = 0.1584$ ns).

Table 3: Multiple-comparisons summary. Primary test family: Tests 1–3 ($k = 3$, $\alpha = 0.05/3$).

Test	N	Raw p	p_{adj}, k=3	Status
Test 1 (full corpus)	1011	0.0027^a	0.008 **	**Primary**
Test 2 (dome A+)	90	0.0401^a (OR = 1.90)	0.015 *	**Primary**
Test 3 (cluster asym.)	1011	0.0002^b	6×10^{-4} ***	**Primary**
Test 4 (temporal)	1011	0.1584^b	—	Descriptive
Test 2b (mound evol.)	110	0.0445^a (OR = 1.79)	N/A	Sensitivity

[a]Fisher exact p vs. rest of corpus (one-sided, enrichment); [b]permutation p. Binomial baselines: Test 2 $p = 0.0049$ **, Test 2b $p = 0.0046$ **.

OR = 0.668). The declining rate may partly reflect UNESCO's expansion into regions where hemispherical traditions are less prevalent. The gradient is treated as descriptive context only (Figure 3).

4.6 FDR audit and Bonferroni correction

Primary and secondary results are consolidated in Table 3.

A comprehensive FDR audit (43 tests; Benjamini & Hochberg 1995) retains 22 at $q < 0.05$, of which 5 survive full Bonferroni at $k = 43$. The 5 full-k survivors are: dome/spherical A++ (binomial $p = 0.0268$); cluster permutation ($p_{\text{adj}} = 0.0086$); harmonic Mann-Whitney ($p_{\text{adj}} = 0.0004$); sequential-peak permutation ($p_{\text{adj}} = 0.028$; §4.5); and evolution combined A++ (binomial $p = 0.0359$).

5 Robustness Checks

5.1 Unit Sensitivity Sweep

Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 10.7 km), the 0.1-beru spacing produces the strongest joint significance across both the dome and full-corpus populations (Online Supplementary Figure S1). The 0.05-beru and 0.20-beru spacings also reach significance, but both are mathematical consequences of the 0.10-beru structure: every 0.10-beru harmonic is also a 0.05-beru harmonic, and 0.20-beru harmonics are a strict subset, so neither constitutes an independent signal. Adjacent non-harmonic spacings (0.06, 0.07, 0.09, 0.11, 0.12 beru) are non-significant in either population, the key discriminating observation. The 0.08-beru spacing reaches nominal dome significance ($p = 0.0330$) but not full-corpus ($p = 0.1956$, ns), reflecting unsystematic local overlap with the 0.10-beru grid rather than a genuine secondary periodicity. Specifically, 6/12 (50.0%) of 0.08-beru dome hits align to 0.10-beru, indicating the significance is due to overlap.

A degree-denominated Monte Carlo sweep (§6.2) confirms this pattern. Candidate intervals at 4° and 8° reach nominal dome significance, but inspection shows that 6 of 7 and all 4 of their hits respectively fall on 12° and 24° nodes shared with the 3° grid, confirming they are consequences of the primary periodicity rather than independent angular units.

5.2 Sensitivity Slope at the Canonical Unit

A fine sweep within ±1% of 0.10 beru (`fine_sweep_audit.py`) shows a ±0.3% shift collapses joint significance in at least one population. A geographic bootstrap of the full cultural corpus finds no permutation in 10000 trials that reproduces the canonical-unit sharp peak at either the A^{+} or A^{++} precision tier ($p = 0.0000$ and $p = 0.0000$ respectively, both ***), confirming 0.10 beru as a specific address. The dome-filter specificity test yields marginal results across all tiers (A^{++}: $p = 0.0503$, $\sim$; A^{+}: $p = 0.1037$, ns; A: $p = 0.0620$, $\sim$), consistent with the test being underpowered at $n = 83$ sites. Restricting to the 22 morphologically unambiguous spherical forms (stupa, tholos) recovers significance at the A tier ($p = 0.0248$, *). The 3° periodicity is independently confirmed by Tests C and D (§4.4).

A log-scale threshold sweep (73 values, 27/28 significant within ±1 log-decade of A+) yields data-optimal bands $A^{++}\approx 0.00149$ beru, $A^{+}\approx 0.00275$ beru, $A\approx 0.00509$ beru (geometric ratio $r^{*}=1.87$); the canonical metrological values (0.0016, 0.00321, 0.00642 beru) are the nearest Babylonian sub-unit boundaries to these optima.

5.3 Global Anchor Sweep

A global sweep (0–360°E, 0.01° resolution, 36,000 trial anchors) assesses whether the $\sim$35°E corridor is unusual globally. The sweep uses the extended corpus described in §3.2 ($N_{\text{ext}} = 1012$), each anchor excluding its own entry, so the focal anchors test against $N = 1011$.

Both anchors place the corridor in the global top 2.0% (36,000 trial anchors, 0–360°E; Gerizim: 88 A+, 98.0$^{\text{th}}$ percentile; Jerusalem: 88 A+, 98.0$^{\text{th}}$ percentile; full table in Online Supplementary Table S4), and the result is anchor-invariant within the 4.1 km corridor window. No other historical meridian reaches significance.

The $x.21$°E periodic structure

Every $x.21$°E longitude (120 anchors) produces identically the maximum A+ count, spaced exactly $3° = 0.1$ beru apart.

The label "$x.21$°E" denotes phase $\approx 2.21°$ mod $3.0°$ (tier-definition-dependent: a wider or narrower A+ threshold shifts this phase). Gerizim's phase (35.269 mod $3.0 = 2.269°$, gap $0.059° \approx 5.6$ km) is within coordinate uncertainty.

Non-independence note. *The 3° periodicity and the unit sensitivity sweep (§5.1) are consistent manifestations of the same underlying structure rather than independent lines of evidence; their primary evidential role is as robustness checks for Tests 2 and 3 (§6.1).*

Formal statistical tests for the $x.21°$ periodicity

The full-corpus Rayleigh ($p_{\mathrm{perm}} = 0.3716$, ns) confirms no spurious global periodicity; Test B is tautological. Tests C ($p_{\mathrm{perm}} = 0.0194$, *) and D ($p_{\mathrm{boot}} = 0.0199$, *) are positive anchor-specific tests; Test E ($p = 0.0830$, †, marginal) is the anchor discovery correction. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) independently confirm 3° as the dominant period without assuming any particular unit.

The dome-periodicity audit (`dome_periodicity_audit.py`) does not distinguish alignment from geographic concentration. Gerizim scores at the 97.3$^{\mathrm{th}}$ percentile among dome sites, dome site phases are approximately uniform ($\chi^2 = 14.89$, df$\,=9$, $p = 0.0940$, trend-level), and all 13 dome A+ sites have phases within $\pm 0.15°$ of the $x.21°$ optimal phase.

Corridor and landmark ranking. Of the 17 A+ sites unique to Gerizim and 17 unique to Jerusalem, 71 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as its own anchor, Gerizim ranks #26 of 1012 (top 2.8%).

5.4 Spatial Autocorrelation

A+ harmonics host 4.45× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC ($\pm 0.5°$, $\rho = 0.110$): DEFF $= 1.495$, $N_{\mathrm{eff}} = 676$, corrected $p = 0.011$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI $[6.69\%, 10.59\%]$ excludes the 6.42% null. Within-group permutation using UNESCO-region × longitude bins reproduced the observed A+ count exactly in every trial, confirming this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.2) address that question directly.

5.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\mathrm{perm}} = 0.095$ †; 1978-1984 canon ($N = 138$) $p = 0.419$ ns; pre-2000 ($N = 501$) $p = 0.136$ ns. (2) Bootstrap from full corpus: dome $p_{\mathrm{boot}} = 0.106$ ns. (3) KDE-smoothed null: expected 64.9 ± 7.8, observed 88, $Z = 2.96$, $p = 0.003$ **. The KDE null ($p = 0.003$ **) confirms dome enrichment; permutation and bootstrap are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.002$ **, bootstrap $p = 0.004$ **). The geographic-concentration nulls (Null A, Null C; §4.2) show that proximity to harmonic longitudes is necessary but not sufficient to explain the dome enrichment.

5.6 Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) independently of the UNESCO corpus (Tenenbaum, 2026). The Kedu Plain (Java) hosts the densest cluster at the 2.5-beru harmonic, anchored by Borobudur (c. 800 CE, 7.23 km, Tier A+) (Miksic, 1990); the World Peace Pagoda at Lumbini achieves A++ precision ($\sim$555 m) uniquely from Gerizim.

Using the identical anchor and null, the corpus yields 42/229 Tier-A sites (18.3%, 1.43× null; binomial p 0.011, *; permutation $Z = 1.00$, $p_{\text{perm}} = 0.149$, ns). No Tier-A+ enrichment is found (binomial $p = 0.102$, ns), consistent with cluster-level rather than individual-site precision. Harmonic nodes with at least one Tier-A site carry 1.73× more stupa sites on average than empty nodes, mirroring the cluster-asymmetry result of Test 3.

Geographic tier breakdown and bimodal structure. All regional p-values below are one-sided Fisher exact tests. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist and Hindu bimodal enrichment in the UNESCO religion audit (§4.1). The Java/Sumatra node (105°–115°E, near the 2.5-beru harmonic at 110.3°E) concentrates 22 of 83 sites at Tier-A (26.5%, OR = 2.27×, p 0.014 *), rising to 22/54 (40.7%, OR = 5.33×, $p < 0.001$ ***) in the 107°–112°E window, where A+ is also elevated at 20.4% (OR = 4.72×, $p = 0.001$ **). The physical heartland, India/Nepal (70°–90°E) and Myanmar/Cambodia (90°–105°E), is A-tier depleted (combined OR = 0.38×, p 0.005 **, one-sided depletion test) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 32.8% (OR = 6.11×, $p < 0.001$ ***) and the combined heartland 25.0% (OR = 8.75×, $p < 0.001$ ***). This A/C anti-correlation mirrors the bimodal pattern in the religion audit (§4.1). Non-heartland Wikidata stupas (outside 70°–110°E, $N = 102$) show the same A-enriched, C-depleted profile: 27.5% reach Tier-A (OR = 3.05×, p 0.001 **) with no C-band enrichment (OR = 0.13×, $p = 1.000$ ns).

5.7 Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 98.0$^{\text{th}}$ percentile; a ±0.3% unit shift collapses joint significance. FDR audit (43 tests) retains 22 at $q < 0.05$; 5 survive Bonferroni at $k = 43$.

6 Limitations

6.1 Circularity: Full Canonical Treatment

Pre-registration cannot address circularity for analyses of permanent public databases. The UNESCO coordinate file and Mount Gerizim are both fully public, so any investigator could run this pipeline at any time and temporal priority is unverifiable. The relevant questions are whether the pattern exists and whether the anchor was chosen on independent grounds.

Both are answered directly. The global anchor sweep (§5.3) confirms that the enrichment is a property of the $\sim$35°E corridor. Every x.21°E anchor produces the same result, and Gerizim and Jerusalem both place the corridor in the global top 2.0% of 36,000 trial anchors. What distinguishes Gerizim is pre-existing documentary evidence, the *tabbur ha-aretz* cosmographic role (§2.3), that predates this analysis entirely.

The 3° periodic structure is recoverable by any unanchored 120-phase scan without reference to Gerizim. The dome enrichment (Test 2) is identical across all 120 x.21°E anchors. The cluster asymmetry (Test 3) requires no keyword filter and no anchor. Test B is tautological by construction (§4.4) and is noted as such.

This ranking is marginal under the discovery correction (Test E, $p = 0.0830$, †); the primary evidence is Tests 2 and 3, which require no anchor.

6.2 Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor of Cultural/Mixed sites. The regional distribution is as follows: (Europe & North America: 50%; Arab States: 9%; Asia-Pacific: 23%; Latin Americas: 11%; Africa: 7%). Domed monuments exceed the A+ rate of other monuments from the same longitude footprint (Null C, $p = 0.0394$, *), establishing that the enrichment is carried by architectural type, not location alone.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacing values from 0.5° to 15°) identifies which angular interval produces clustering beyond geographic concentration without assuming any particular unit. The dominant peak is at 3° ($p < 0.001$), with harmonics at 6° and 12° also significant. Collapsing harmonics, the dome subset yields 1 independent dominant spacings ($P = 0.1951$ under the geographic null), supporting 3° as the period to which domed monuments are preferentially aligned.

6.3 Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows a modestly lower A+ rate (5.5%) than the 6.42% null. The one-sided binomial depletion test gives $p = 0.4102$ ns, but the result is suggestive rather than definitive.

6.4 Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids, introducing unsystematic noise but not directional bias.

Metrological evidence. No physical surveying instrument calibrated in beru subdivisions has been recovered; documentary sources attest the beru system (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). This study therefore treats the coordinate alignment of Mount Gerizim as a novel empirical observation.

7 Discussion

7.1 Alternative Explanations

(1) Independent convergence. Hemispherical structures are a recurring architectural solution across cultures (Aveni, 2001, 2003), but independent convergence predicts no longitude specificity.

(2) Geographic concentration. Null C directly tests this: other monuments from the same longitude footprint are significantly less enriched ($p = 0.0394$, *), establishing that the signal is carried by architectural type, not location. The Monte Carlo unit sweep (§6.2) independently confirms 3° as the dominant angular period without assuming any unit.

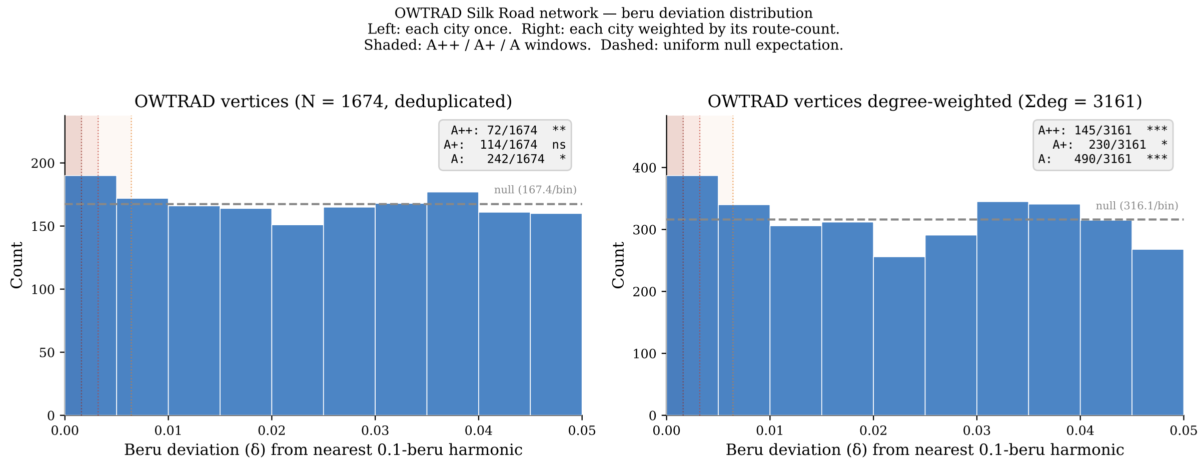

Figure 4: OWTRAD Silk Road endpoint tier distributions: 1674 deduplicated endpoints (left) and 3498 degree-weighted positions (right). Both show significant A++ enrichment (deduplicated: $z = 2.56$, $p = 0.0084$ **; degree-weighted: $z = 4.23$, $p =< 0.0001$ ***).

(3) Random coincidence. The KDE-smoothed null confirms dome enrichment ($p = 0.003$ **); permutation ($p = 0.095$ †) and bootstrap ($p = 0.106$ ns) are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.002$ **; bootstrap $p = 0.004$ **). Tests C and D (§4.4) confirm anchor specificity and periodicity structure.

7.2 Architectural Stratification and Route-Segment Alignment

The enrichment signal stratifies by architectural type and religious tradition (Tests 2/2b, §4.2–4.3; §4.1), with domed monuments and the Buddhist/Hindu sub-groups showing the strongest harmonic concentration. The bimodal distribution in both traditions is consistent with the 0.1-beru grid producing discrete attractor positions rather than a continuous distribution.

An independent check on route geography comes from the OWTRAD attested-route network (Ciolek, 2004) (1946 route segments, 1674 unique endpoints). Testing all 3498 endpoint positions across the full edge list (each city weighted by the number of segments it connects), 156 fall at A++ harmonics, 250 at A+, and 530 at A (1.39× null rate, $z = 4.23$, $p =< 0.0001$ *** for A++; $p = 0.0445$ * for A+; $p =< 0.0001$ *** for A; one-tailed binomial). Testing each unique endpoint once gives 72/1674 at A++, 114 at A+, and 242 at A (1.34×, $z = 2.56$, $p = 0.0084$ ** for A++; $p = 0.2708$ ns for A+; $p = 0.0276$ * for A; one-tailed binomial). A phase-shift test (random offset of the 3° grid, 100000 iterations) gives $p = 0.0108$ *, confirming the enrichment is specific to the Gerizim anchor. A cluster asymmetry test mirrors Test 3 of the primary family: harmonics containing at least one A++ endpoint (25 harmonics) carry 2.40× more total route-connectivity than harmonics without one (16 harmonics; $p = 0.0003$ ***, Mann-Whitney one-tailed). The Silk Road corridor is illustrated in Figure 4; harmonic tier distributions for both OWTRAD populations are shown in Figure **??**.

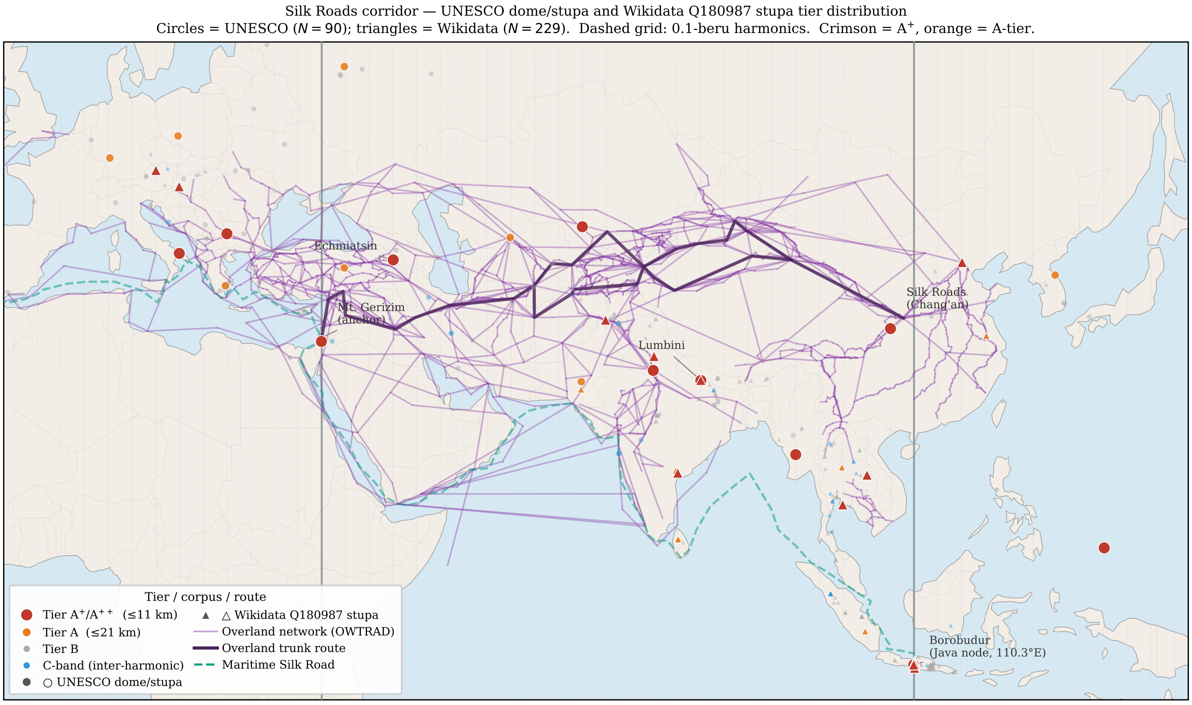

Figure 5: Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor. Overland route edges from the OWTRAD attested-route database (Ciolek, 2004). Maritime route is a schematic coastal polyline after Casson (1989), Sen (2014), and Tomber (2008). An interactive version is provided as supplementary material (`supplementary/interactive_map/silkroad_interactive.html`).

8 Conclusion

The UNESCO Cultural and Mixed corpus shows significant enrichment at harmonics of the Babylonian beru anchored at Gerizim, with all three primary tests surviving Bonferroni correction at $k = 3$. Longitude bands containing a precisely aligned monument are 4.45× more densely populated with other heritage sites than bands without one. Precise alignment is most concentrated among the earliest-dated sites and declines in later cohorts, consistent with a founding canon. The 3-degree periodicity is recoverable without specifying an anchor in advance; Gerizim has independent documentary support as a sacred mountain. The Q180987 stupa corpus and the OWTRAD route network each reproduce the harmonic independently (§5.6).

Geographically, Mount Gerizim lies between the Mediterranean and the overland Silk Road (Hansen, 2012), placing it at a natural node of corridors through which architectural canons circulated in antiquity (Figure 4).

A Keyword Audit Summary

Full site-by-site audit files covering keyword classification, tier listings, anchor sweep, corridor precision, unit sweep, stupa geography, and FDR output are provided in `supplementary/audit/` in the archived repository (Tenenbaum 2026); all regenerable from scripts in `tools/`.

A.1 Dome/Spherical Monument Filter (Tests 2, 2b, 2x, 2bx)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text. All 90 matching sites are included in Test 2 without context gating.

For Test 2x, ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The seven rejected sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuṭa, and Diquís Stone Spheres.

Test 2b adds earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 2bx applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.2 and §4.3.

A.2 Founding-Origin Classifier (Test 5, Post-Hoc)

Results. The classifier finds 33.0% of A+ sites carry a founding keyword versus 28.7% in the full corpus (Fisher OR = 1.25, $p = 0.209$ ns). The A++ sub-population (44 sites) shows a stronger signal: 47.7% carry a founding keyword (OR = 2.37, $p = 0.005$ **), consistent with the most precisely aligned sites being disproportionately historically foundational monuments.

The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum, 2026).

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Data Availability Statement

All data and analysis code are openly available. A permanent archival snapshot (v1.3.0) is deposited at Zenodo (Tenenbaum 2026; https://doi.org/10.5281/zenodo.19868418). The UNESCO World Heritage XML dataset version used in this study is preserved in the Zenodo archive; the primary source is UNESCO World Heritage Centre 2024.

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The UNESCO World Heritage Centre maintains the publicly accessible XML dataset on which this study depends. This research was conducted independently and without external institutional affiliation.

Conflict of Interest

The author declares no conflict of interest.

Software and AI Disclosure

All analyses were performed in Python 3 with NumPy, SciPy, and pandas. Claude (Anthropic) assisted with manuscript drafting and prose editing. GitHub Copilot (Microsoft) assisted with code autocompletion. All scientific conclusions, hypothesis design, keyword-list decisions, and analytical choices are solely the author's.